

表 9: 不同模型在多个数据集上的异常检测结果

| 模型                 | FlowGraph       |                 |                 |                 |                 |                 |                  | TraceLog        |                 |                 |                 |                 |                 |                 | ADFA-LD         |                 |                 |                 |                 |                 |                 | HDFS            |                 |                 |                 |                 |                 |                 |
|--------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|                    | AUROC           | AP              | P@1%            | R@1%            | TPR@1%FPR       | F1              | MCC              | AUROC           | AP              | P@1%            | R@1%            | TPR@1%FPR       | F1              | MCC             | AUROC           | AP              | P@1%            | R@1%            | TPR@1%FPR       | F1              | MCC             | AUROC           | AP              | P@1%            | R@1%            | TPR@1%FPR       | F1              | MCC             |
| DeepTraLog         | 0.9785 ± 0.0228 | 0.9856 ± 0.0135 | 1.0000 ± 0.0000 | 0.0300 ± 0.0000 | 0.9667 ± 0.0289 | 0.9796 ± 0.0136 | 0.9644 ± 0.0235  | 0.6472 ± 0.0163 | 0.5416 ± 0.0430 | 0.6667 ± 0.5774 | 0.0193 ± 0.0167 | 0.0694 ± 0.0673 | 0.1228 ± 0.1171 | 0.1210 ± 0.1666 | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               |
| DeepTraLog-adapted | –               | –               | –               | –               | –               | –               | –                | –               | –               | –               | –               | –               | –               | –               | 0.7608 ± 0.0038 | 0.4394 ± 0.0061 | 0.6667 ± 0.1000 | 0.0268 ± 0.0040 | 0.0393 ± 0.0015 | 0.2270 ± 0.0412 | 0.1038 ± 0.0425 | 0.7379 ± 0.0120 | 0.6389 ± 0.0035 | 1.0000 ± 0.0000 | 0.0607 ± 0.0000 | 0.4941 ± 0.0018 | 0.6440 ± 0.0035 | 0.6349 ± 0.0051 |
| GLocalKD           | 0.9561 ± 0.0082 | 0.9738 ± 0.0025 | 1.0000 ± 0.0000 | 0.0300 ± 0.0000 | 0.9500 ± 0.0000 | 0.9727 ± 0.0029 | 0.9526 ± 0.0054  | 0.6795 ± 0.0060 | 0.6318 ± 0.0057 | 1.0000 ± 0.0000 | 0.0289 ± 0.0000 | 0.2912 ± 0.0089 | 0.4388 ± 0.0152 | 0.4089 ± 0.0278 | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               |
| GLocalKD-adapted   | –               | –               | –               | –               | –               | –               | –                | –               | –               | –               | –               | –               | –               | –               | 0.7236 ± 0.0111 | 0.4164 ± 0.0134 | 0.6556 ± 0.3372 | 0.0264 ± 0.0136 | 0.0290 ± 0.0182 | 0.2503 ± 0.0229 | 0.1635 ± 0.0155 | 0.7559 ± 0.0057 | 0.6467 ± 0.0019 | 1.0000 ± 0.0000 | 0.0607 ± 0.0000 | 0.4963 ± 0.0018 | 0.6450 ± 0.0008 | 0.6348 ± 0.0000 |
| HGT                | 0.5374 ± 0.4560 | 0.4960 ± 0.1967 | 0.0000 ± 0.0000 | 0.0000 ± 0.0000 | 0.0000 ± 0.0000 | 0.0000 ± 0.0000 | −0.0693 ± 0.0745 | 0.6583 ± 0.0582 | 0.5532 ± 0.0662 | 0.5833 ± 0.3643 | 0.0169 ± 0.0105 | 0.1340 ± 0.1127 | 0.2277 ± 0.1195 | 0.2557 ± 0.1014 | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               |
| HRA-GNN            | 0.6646 ± 0.5489 | 0.7462 ± 0.4131 | 0.6667 ± 0.5774 | 0.0200 ± 0.0173 | 0.6333 ± 0.5485 | 0.6463 ± 0.5597 | 0.6110 ± 0.5810  | 0.8150 ± 0.0123 | 0.7497 ± 0.0225 | 1.0000 ± 0.0000 | 0.0289 ± 0.0000 | 0.2969 ± 0.0675 | 0.4482 ± 0.0790 | 0.4341 ± 0.0663 | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               |
| HRGCN              | 0.9333 ± 0.1155 | 0.8672 ± 0.2300 | 0.6667 ± 0.5774 | 0.0200 ± 0.0173 | 0.6667 ± 0.5774 | 0.6650 ± 0.5759 | 0.6438 ± 0.6093  | 0.7381 ± 0.0441 | 0.6961 ± 0.0321 | 1.0000 ± 0.0000 | 0.0289 ± 0.0000 | 0.3365 ± 0.0051 | 0.4795 ± 0.0318 | 0.4658 ± 0.0277 | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               | –               |
| OCHetGCN           | 0.9557 ± 0.0561 | 0.8898 ± 0.1494 | 0.6667 ± 0.5774 | 0.0200 ± 0.0173 | 0.4200 ± 0.5188 | 0.5299 ± 0.5007 | 0.4754 ± 0.5257  | 0.7062 ± 0.0603 | 0.6015 ± 0.0551 | 0.6917 ± 0.2898 | 0.0200 ± 0.0084 | 0.1015 ± 0.1302 | 0.2167 ± 0.1662 | 0.2296 ± 0.1674 | 0.7938 ± 0.0030 | 0.5092 ± 0.0029 | 0.1444 ± 0.0509 | 0.0058 ± 0.0020 | 0.0027 ± 0.0000 | 0.5154 ± 0.0117 | 0.3973 ± 0.0108 | 0.7368 ± 0.0124 | 0.6380 ± 0.0101 | 1.0000 ± 0.0000 | 0.0607 ± 0.0000 | 0.4973 ± 0.0018 | 0.6467 ± 0.0051 | 0.6411 ± 0.0111 |